



Department of CSE

Mini Project Report - Task 1

“Malware Family Classification Using an Image-to-Graph GCN on the MaleBin Dataset”

Course Code: ICE 478/CSE475

Section: 2

Track: Track 2 - GCN (Image -> Graph)

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1. Introduction

Malware is a major cybersecurity threat because new variants are continually developed to evade traditional signature-based detection. Image-based malware classification offers a practical alternative by converting binary content into grayscale images, allowing machine learning models to learn recurring visual and structural patterns.

This project uses the MaleBin Malware Binary Greyscale Images dataset for multi-class malware family classification and follows Track 2, where each image is converted into a graph before graph convolution is applied. The exploratory analysis in Task 1 is therefore used not only to understand the image data, but also to judge how a patch- or region-based graph can be constructed later. Because malware byte-images are texturelike rather than natural photographs, the graph structure must be treated as an experimental representation rather than a naturally occurring semantic region graph.

2. Dataset Overview

The MaleBin dataset contains grayscale visual representations of malware binaries. Images are arranged by malware family, which supports supervised multi-class classification. The Task 1 analysis identified the following dataset characteristics.

Property	Value
Dataset	MaleBin Malware Binary Greyscale Images
Source	Kaggle - tashiee/malebin-malware-binary-greyscale-images
Application area	Malware family classification
Total images	12,464
Number of classes	39
Image format	PNG/JPG grayscale
Observed balance	Moderate class imbalance
Observed resolution	256 x 256 in the inspected dimension sample

3. Exploratory Data Analysis (EDA)

The analysis covered dataset organization, visual appearance, class balance, dimensions, pixel intensity, image metadata, statistical pixel features, correlation, and low-dimensional structure. These checks establish the preprocessing requirements and provide evidence for later graph construction.

3.1 Dataset Description

Each malware family is stored in a separate class folder. This structure makes the folder name a direct class label during supervised loading. The dataset scan confirmed that images could be indexed by family and prepared for later train, validation, and test splitting.

	Property	Value
0	Dataset name	MaleBin (Malware Binary Greyscale Images)
1	Source	Kaggle (tashiee/malebin-malware-binary-greysca...
2	Application area	Malware family classification from binary-to-i...
3	Total images	12464
4	Number of classes	39
5	Image format	PNG/JPG greyscale
6	Balanced?	TBD — see class balance below

Figure 3.1: Dataset Summary Extracted During EDA

3.2 Sample Malware Images

Random samples from several malware families show visibly different byte-image textures. Some families contain dense repetitive bands, while others contain more irregular or relatively uniform regions. These differences indicate that local texture and spatial arrangement can provide useful classification information.

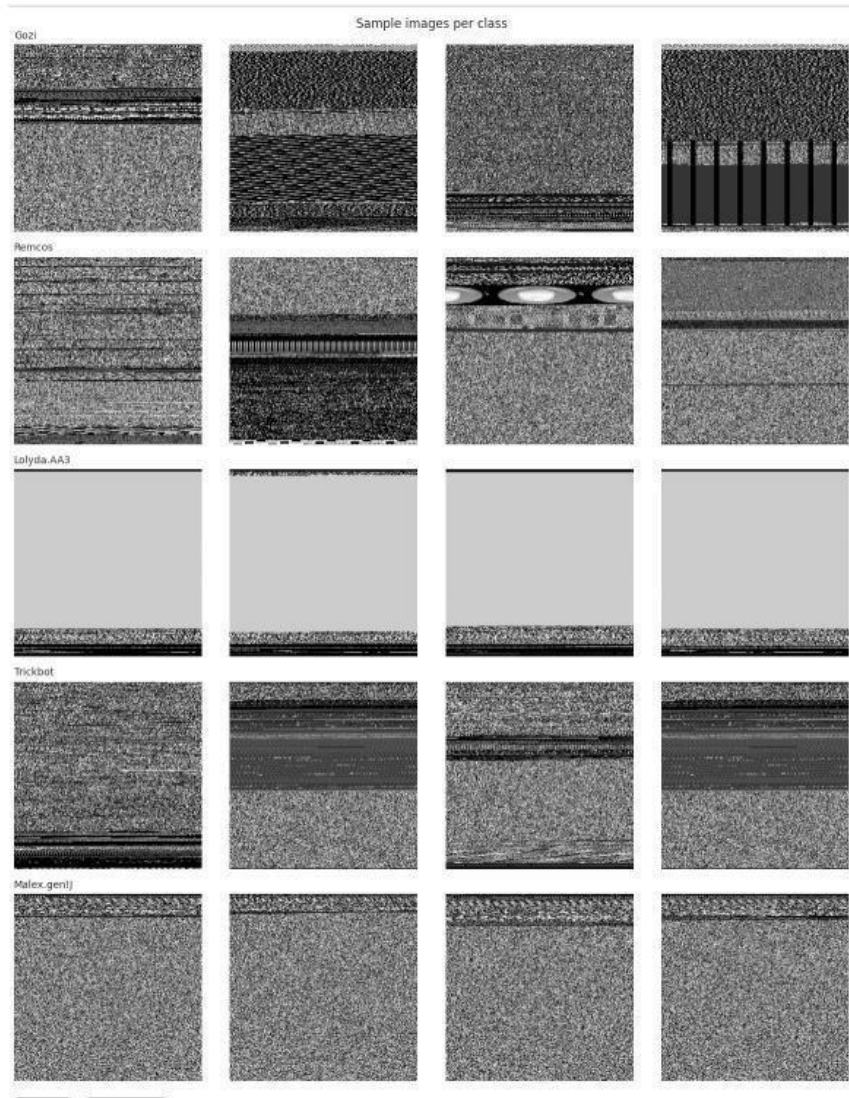


Figure 3.2: Sample Malware Images from Different Families

3.3 Class Distribution

The class-count analysis shows a moderate imbalance. A few families contain close to one thousand images, while several minority families contain substantially fewer samples. This makes accuracy alone insufficient and supports the later use of class-aware sampling and macro-averaged metrics.

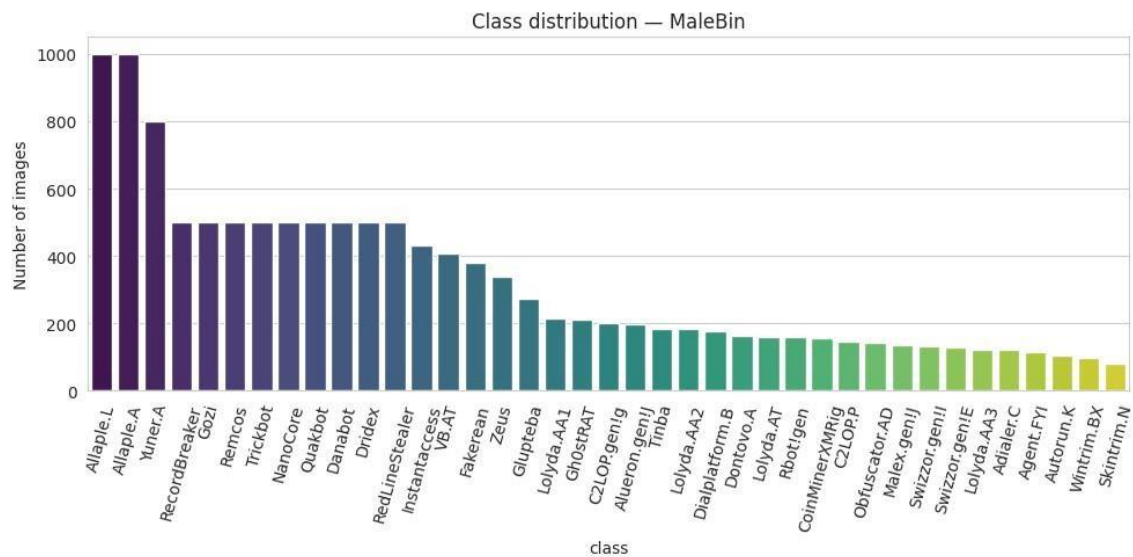


Figure 3.3: Class Distribution of Malware Families

3.4 Image Dimension Analysis

The dimension check shown below reports 2,000 inspected images, all with width and height equal to 256 pixels. Therefore, the analyzed subset is already spatially consistent. Later models may still resize images to match a selected CNN or graph-construction resolution, but resizing is a modeling choice rather than a correction for inconsistent dimensions.

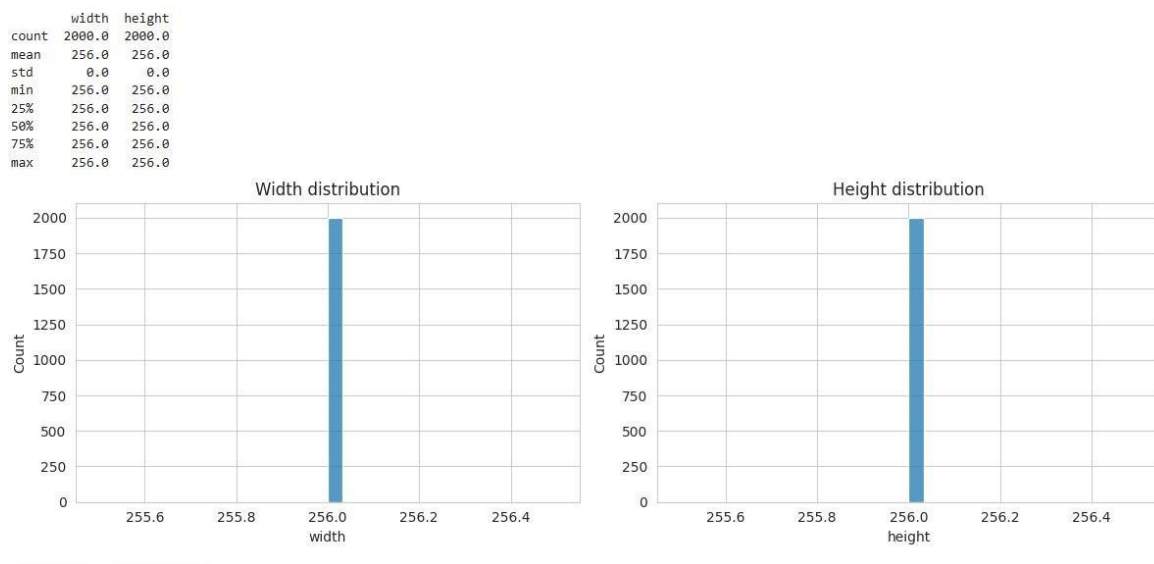


Figure 3.4: Distribution of Image Dimensions

3.5 Pixel Intensity Distribution

Pixel values span the grayscale range from 0 to 255. The distribution is concentrated strongly near dark values, with additional structure across the mid and high intensity ranges. Normalization is therefore appropriate before neural-network training so that input magnitude is controlled consistently.

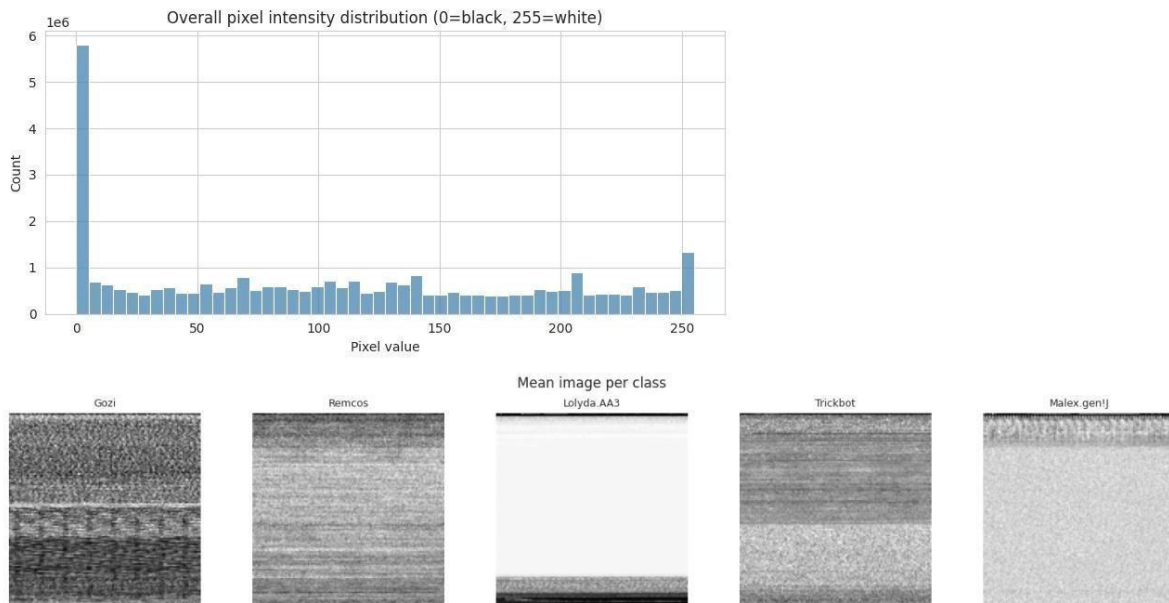


Figure 3.5: Pixel Intensity Distribution of the MaleBin Dataset

3.6 Image Metadata Analysis

Metadata inspection considered image mode, width, height, file size, aspect ratio, and labels. The observed samples are grayscale and structurally consistent, supporting automated loading and fixed-grid patch extraction in the later GCN pipeline.

label	width	height	aspect_ratio	mode	pixels	filename	file_size_kb	label_id	cnn_ready	suggested_size	ni
Malware_Merge/Adialer.C	256	256	1.0	L	65536	0528185d9960c8bd1b6692d2ccc8b9dd.png	43.902344	0	True	224x224	
Malware_Merge/Adialer.C	256	256	1.0	L	65536	050f962b76f3b115044873d9e3a07a97.png	43.903320	0	True	224x224	
Malware_Merge/Adialer.C	256	256	1.0	L	65536	053f41978f89e436244c691072f4a548.png	43.495117	0	True	224x224	
Malware_Merge/Adialer.C	256	256	1.0	L	65536	06e21af7e2fdd67622d0122e113b0d53.png	43.627930	0	True	224x224	

Figure 3.6: Summary of Image Metadata

3.7 Pixel Feature Distribution

Statistical image descriptors such as mean intensity, standard deviation, entropy, and bright/dark pixel ratios vary across malware families. These distributions demonstrate that local patches can carry different intensity and texture information, which motivates using patch-level features as graph-node attributes.

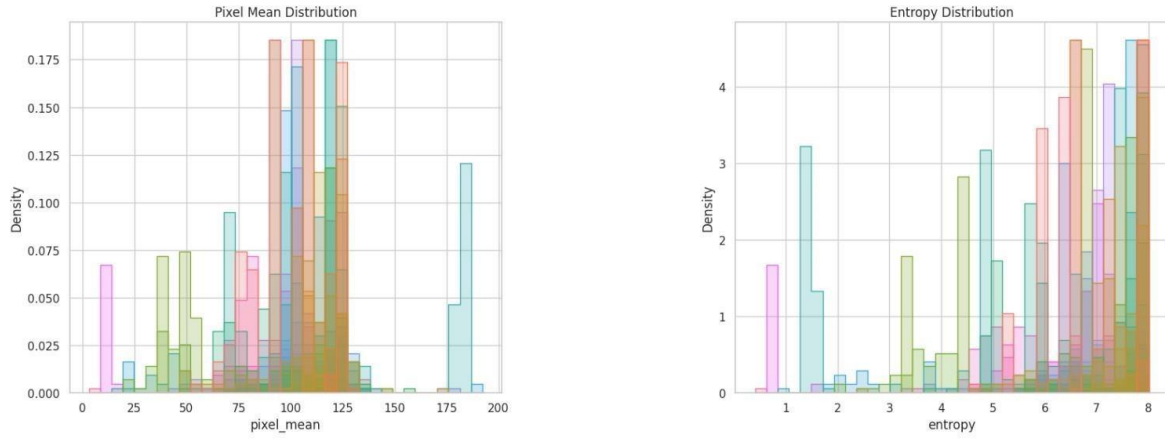


Figure 3.7: Distribution of Extracted Pixel Features

3.8 Correlation Heatmap of Pixel Features

The correlation matrix reveals strong relationships among several simple image statistics. For example, pixel mean is strongly negatively related to the dark-pixel ratio, while bright-pixel ratio and entropy are positively related. This indicates that some handcrafted features are redundant, which later motivates richer learned CNN features for the improved graph model.

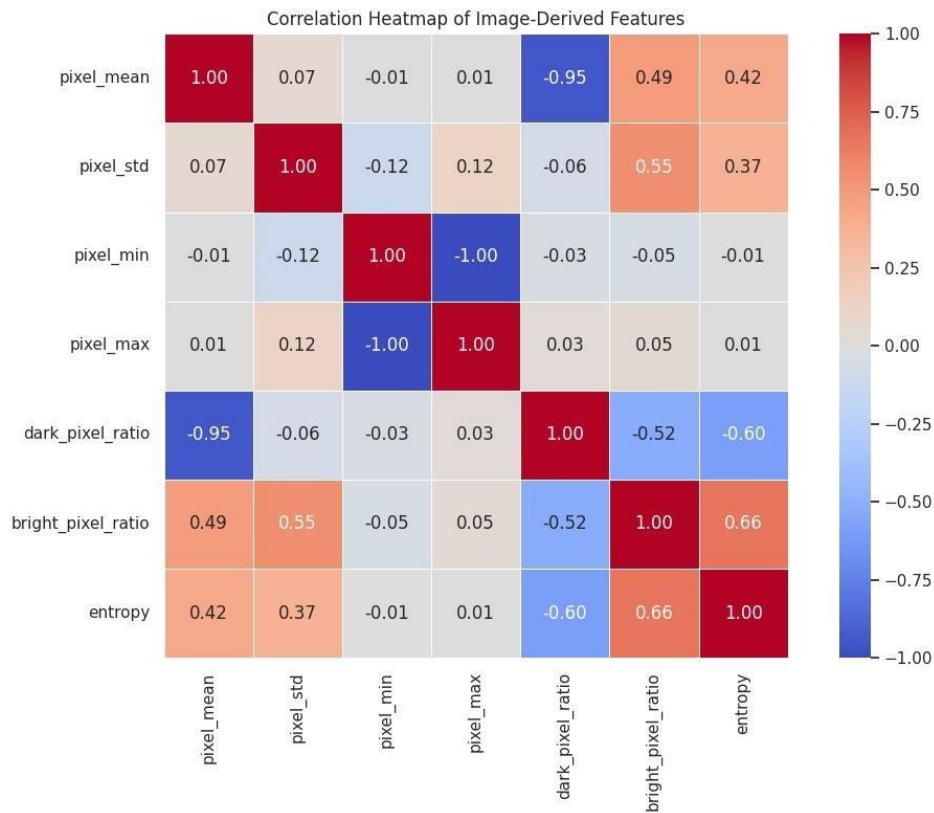


Figure 3.8: Correlation Heatmap of Pixel Features

3.9 Principal Component Analysis (PCA)

PCA was applied to the extracted statistical features. The projection shows heavy overlap among many classes, although some samples and families occupy distinct regions. The overlap suggests that a small set of global statistics is not sufficient for high-quality family separation, so spatially localized image features are needed.

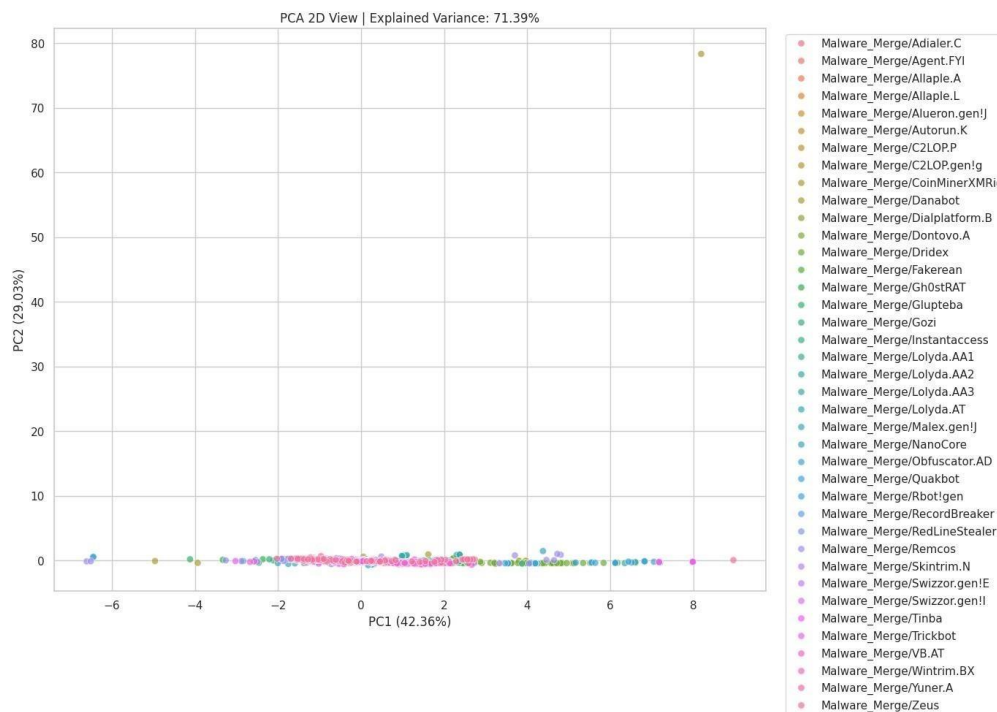


Figure 3.9: PCA Projection of Malware Samples

4. Key Observations and Track 2 Implications

- The dataset contains 12,464 malware images distributed across 39 families.
- Different families exhibit visibly different grayscale texture patterns.
- Class frequencies are moderately imbalanced, so macro-F1 and per-class recall are important evaluation measures.
- The inspected image-dimension sample is consistently 256 x 256.
- Pixel intensities are dominated by dark and mid-tone values, supporting input normalization.
- Simple statistical features show correlation and substantial class overlap in PCA.
- For Track 2, a patch graph is more defensible than claiming semantic superpixels, because malware byteimages do not contain natural objects or lesions.
- Graph construction should be performed only after the dataset split to avoid leakage between training and evaluation data.

5. Conclusion

The Task 1 EDA shows that MaleBin is suitable for image-based malware family classification and provides enough local texture variation to test an image-to-graph model. The data are organized by family, moderately

imbalanced, and visually structured, while simple global pixel statistics alone do not fully separate the classes. For the Track 2 pipeline, the next stage will compare conventional CNN baselines with a patch-based GCN in which local image regions become graph nodes and spatial relationships form graph edges. The graph representation will be evaluated honestly because byte-image regions do not have the same semantic meaning as regions in natural or medical images.